

# Cloud Repatriation: Migrating Public Cloud FaaS to Hybrid Architecture

# Introduction & Motivation

Lecture: Cloud Repatriation: Migrating Public Cloud FaaS to Hybrid Architecture

# What is Cloud Repatriation?

- Repatriation = people returning to their country of origin (patria = homeland, from Latin). More generally, bringing something to its origin, home, previous state or place.
- In case of cloud repatriation, this does not necessarily *mean returning from the cloud*, but more generally, *moving away from the cloud*
- Thus, **cloud repatriation = migrating workloads, data or applications from (public) cloud to on-premises or hybrid environments**

# Why migrate away from the Cloud?

- Rising cloud costs or challenges in estimating future costs
- Regulations (e.g., GDPR, DORA, data locality) and security concerns
- Latency and performance bottlenecks
- Lock-in and vendor-specific limitations – who has the control?

# Before & After

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# Migration – Before vs. After

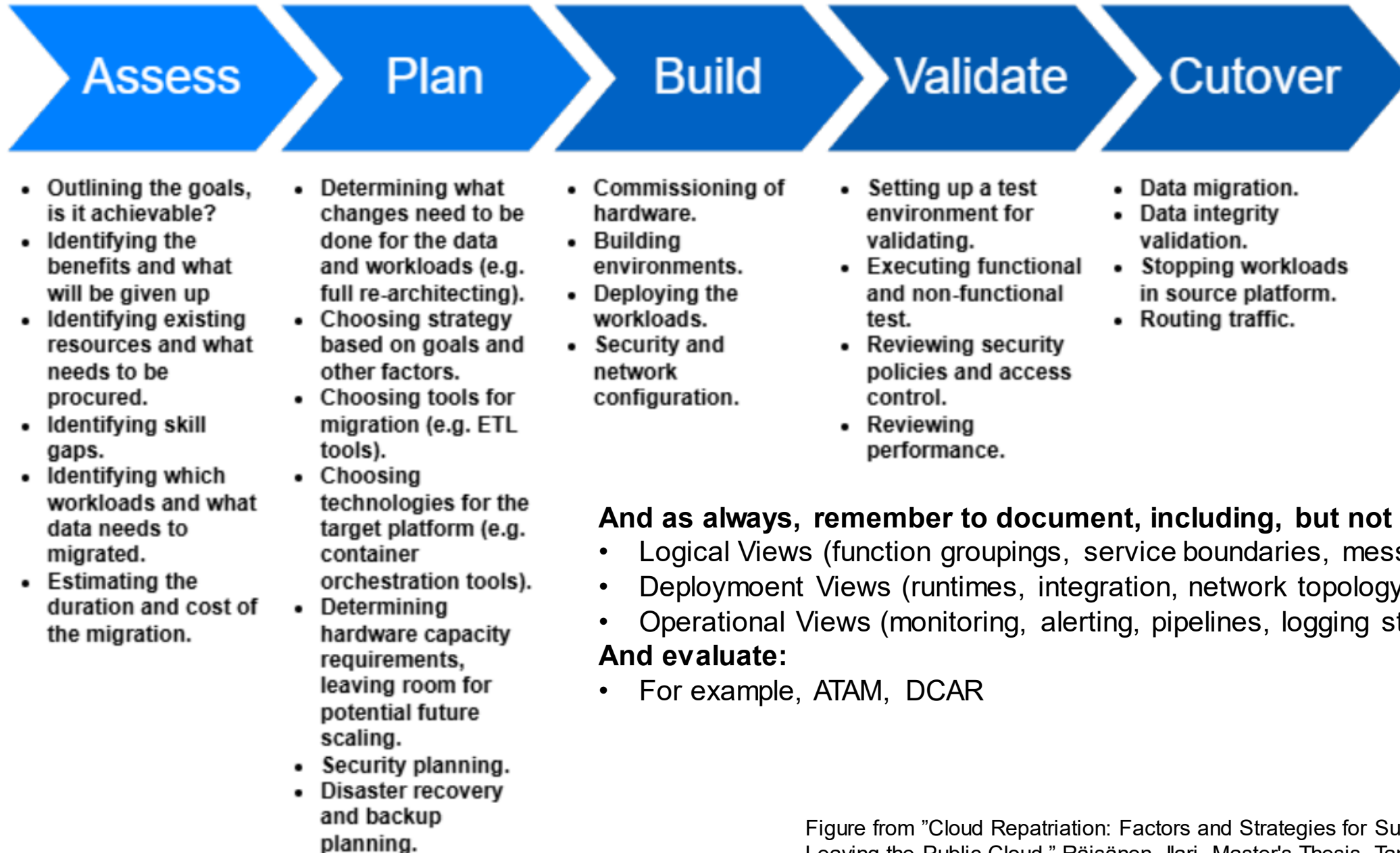
## Public Cloud

- Cloud-native services
  - E.g., AWS Lambda, Azure Functions, Google Run Functions
- Implicit scalability, monitoring, integration
- Tight integration with other cloud services
- Less control, but simpler deployment, operation and development

## Hybrid or On-Prem

- Custom infrastructure
  - E.g., OpenFaaS, Knative, Kubernetes
- Self-managed scaling and observation
- (Re-)integration with internal systems, edge devices and services
- More control, but more complex CI/CD

# Migration – Repatriation Timeline



# Summary

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# Summary

- Cloud repatriation is rising due to cost, compliance, and architecture control needs.
- Migrating FaaS to hybrid/on-prem requires rethinking deployment, operations, and architectural style.
- Architectural documentation (views/models) remains essential in hybrid environments.
- Evaluation methods can help assess migration decisions and trade-offs.

# References & More to Read

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